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Diseñe la armadura de la viga 25/50 mostrada en la figura .Considere hormigón G25, acero A630-420H y un recubrimiento de 5 cm. Datos: $P_L := 4\ \text{tonnef}$ $q_L := 800\ \frac{\text{kgf}}{\text{m}}$

Nota : No olvide considerar el peso propio de la viga

Viga simplemente apoyada con carga distribuida y una carga puntual al centro de la luz

Datos

$b := 25\ \text{cm}$ $h := 50\ \text{cm}$ $L := 6\ \text{m}$ $\gamma_c := 2500\ \frac{\text{kgf}}{\text{m}^3}$

$r := 5\ \text{cm}$ $d := h - r = 45\ \text{cm}$

$f_c := 250\ \frac{\text{kgf}}{\text{cm}^2}$ $f_y := 4200\ \frac{\text{kgf}}{\text{cm}^2}$ $\beta := 0.85$ $E_s := 2100000\ \frac{\text{kgf}}{\text{cm}^2}$

Solución

$q_{pp} := \gamma_c \cdot b \cdot h = 0.3125\ \frac{\text{tonnef}}{\text{m}}$

$q_D := q_{pp}$

$M_{u1} := \frac{1}{8} \cdot 1.4 \cdot q_D \cdot L^2 = 1.9688\ \text{tonnef m}$ $U_1 := 1.4 \cdot D$

$M_{u2} := \frac{1}{8} \cdot 1.2 \cdot q_D \cdot L^2 + \frac{1}{8} \cdot 1.6 \cdot q_L \cdot L^2 + 1.6 \cdot P_L \cdot \frac{L}{4} = 17.0475\ \text{tonnef m}$ $U_2 := 1.2 \cdot D + 1.6 \cdot L$

$M_u := \max\left(\begin{bmatrix} M_{u1} \\ M_{u2} \end{bmatrix}\right) = 17.0475\ \text{m tonnef}$

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Condición de diseño

$\phi M_n := M_u \qquad \phi := 0.9$

$$\phi M_n := \phi \cdot \rho \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right)$$

$$\rho := \text{polyroots}\left(\begin{bmatrix} -M_u \\ \phi \cdot b \cdot d^2 \cdot f_y \\ (-\phi) \cdot b \cdot d^2 \cdot f_y \cdot 0.59 \cdot \frac{f_y}{f_c} \end{bmatrix}\right) = \begin{bmatrix} 0.0099 \\ 0.091 \end{bmatrix}$$

$\rho := 0.0099$

$\rho_{max} := 0.32 \cdot \beta \cdot \frac{f_c}{f_y} = 0.0162 \qquad \rho < \rho_{max}$ Cumple la norma

$A_s := \rho \cdot b \cdot d = 11.1375\ \text{cm}^2$ (3ϕ22) (10 ϕ12)

Armadura inferior

3 barras de diametro 22 ($A_s := 11.4\ \text{cm}^2$)

$n_{filas} := 1 \qquad n_b := 3 \qquad \phi_b := 22\ \text{mm} \qquad \phi_s := 8\ \text{mm} \qquad r_{est} := 3\ \text{cm}$ (al borde libre lateral del estribo)

$$s := \frac{b - 2 \cdot r_{est} - \frac{n_b}{n_{filas}} \cdot \phi_b - 2 \cdot \phi_s}{\frac{n_b}{n_{filas}} - 1} = 5.4\ \text{cm} \qquad s_{min} := \max\left(\begin{bmatrix} 25\ \text{mm} \\ \phi_b \end{bmatrix}\right) = 2.5\ \text{cm}$$

$s > s_{max}$ ok

$$\phi M_n := \phi \cdot \rho \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right) = 17.0858\ \text{tonnef m}$$

$FU := \frac{M_u}{\phi M_n} = 0.9978 \qquad ok$

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